“BAYMAX: Health Advisor”

Submitted in partial fulfillment of the requirements

For the degree of

Bachelor of Technology

Computer Science & Engineering Department

Submitted By

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Under the Guidance of

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AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY

AMITY UNIVERSITY MUMBAI

2025-2026

Certificate

This is to certify that the mini-project entitled **"BAYMAX: Health Advisor"** is a bonafied work of Arisha Khan, Shreya Dilipkumar, Parvathikrishna and Avani Mittal submitted to the Amity School of Engineering and Technology, Amity University Mumbai in partial fulfilment of the requirement for the degree of Bachelor of Technology .

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Approval

I declare that this written submission conveys my ideas in my own words. I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not mispresented or fabricated or falsified any idea/date/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the institute and the can evoke penal action form the sources which have thus not been properly cited or from whom proper permission has not been taken when needed

Examiners

Dr. Deepa Parasar

Date

Place

Declaration of Academic Integrity

I declare that this written submission conveys my ideas in my own words. I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not mispresented or fabricated or falsified any idea/date/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the institute and the can evoke penal action form the sources which have thus not been properly cited or from whom proper permission has not been taken when needed

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Chapter 1
Introduction

The growing use of digital health-support systems has created a strong need for structured and reliable datasets that help with early condition identification and general health awareness. A well-designed dataset ensures that symptoms, lifestyle patterns, and demographic details are organised clearly so they can be interpreted in a meaningful way. Such structured information forms the base for generating observations, reducing ambiguity, and offering users a simple and supportive healthmonitoring experience. By including symptom indicators along with basic lifestyle and demographic factors, the dataset becomes useful for developing systems that aim to provide organised and user-friendly health guidance.

The motivation behind this project is to create a dataset that is simple, accessible, and suitable for generating preliminary health observations. The dataset brings together multiple factors such as age, gender, stress levels, sleep duration, and hydration to give a more complete view of general well-being. Its scope includes covering common symptoms across respiratory, gastrointestinal, neurological, urinary, allergic, and general categories, supported by relevant lifestyle inputs. The main objective is to prepare a privacy-safe and systematic dataset that contains no real personal information, ensuring safe usage while enabling tasks like data validation, pattern identification, and condition summarisation. This chapter outlines the purpose, structure, and value of the dataset in creating a reliable foundation for digital healthsupport platforms.

Chapter 2
Literature Survey

The importance of structured datasets in health-related and decision-support systems has been widely discussed in various research studies. Many authors highlight that the reliability of any guidance or recommendation platform begins with how well the underlying information is collected, organised, and represented. In digital health environments, where users expect clear and meaningful interpretations of their symptoms, the role of properly formatted datasets becomes even more significant. Research consistently shows that the quality of input information directly influences the clarity and usefulness of the outputs. This demonstrates that before any system can provide observations or suggestions, it must be supported by a dataset that is organised, complete, and aligned with the intended application.

Several researchers have emphasised the need for domain-specific datasets that combine multiple types of features. Studies in agriculture, healthcare, and general decision systems show a similar pattern dataset design must include a clear selection of attributes that contribute to meaningful interpretation. For instance, Suresh et al. examined how structured parameters lead to better identification patterns in digital platforms. Their work demonstrated that well-chosen features and organised data representation form the base for consistent analysis. This idea applies equally to health datasets, where symptoms, demographic details, and lifestyle indicators must be arranged in a systematic manner so that they can support accurate categorisation.

Another relevant study by Anantha Reddy and colleagues explored decision-making using datasets that capture environmental and physiological factors. Their research highlighted how the combination of multiple attributes—when organised correctly helps systems deliver clearer guidance to users. They also stressed the importance

of ensuring that datasets do not contain missing or contradictory values, because inconsistencies can lead to confusion and unreliable results. This perspective supports the need for structured symptom-related inputs that can be processed without conflict or ambiguity.

Satish Babu discussed how advisory models depend heavily on well-prepared data, especially when assisting users who rely on simplified digital tools. His work explained that datasets should be designed to reflect real-life conditions as closely as possible within a controlled structure. This ensures that users receive insights that match their actual situations. Although his study focused on precision agriculture, the underlying principle is directly applicable in the health domain: datasets must be organised in a way that addresses the user's needs while maintaining clarity, consistency, and reliability.

Research focusing on information diversity, such as the work by Liying Yang, further stresses that datasets benefit from including features that represent different aspects of the user's condition. Yang pointed out that when multiple types of attributes are included such as symptoms, habits, and environmental factors—the resulting dataset becomes more useful in generating well-rounded interpretations. This aligns with the idea of combining symptom indicators with lifestyle patterns like sleep levels, hydration habits, and stress conditions. A dataset that incorporates diverse yet relevant factors provides a richer understanding of the user's state and helps systems produce more complete observations.

Another important contribution comes from Paul et al., who studied structured data in the context of predicting behavioural outcomes. They found that a clear categorisation of attributes makes it easier to group individuals into meaningful classes.

Their research showed that when datasets are divided into logical categories—such as symptoms, demographics, lifestyle, and outputs—the processing becomes more organised and the results more understandable. This supports the structure used in health-related datasets where different feature groups work together to create a complete picture of an individual's condition.

Across all these studies, one common message is repeated: datasets designed for user-facing systems must prioritise clarity, consistency, and completeness. Whether the domain is health, agriculture, or any type of digital guidance platform, the principles remain the same. Researchers consistently emphasise the need to choose relevant features, avoid unnecessary complexity, and structure the dataset in a way that supports practical interpretation. This literature strongly supports the construction of a symptom-based dataset that includes binary indicators, demographic details, lifestyle values, and well-defined condition categories. It shows that the broader research community acknowledges the value of such structured datasets in creating reliable, meaningful, and user-friendly systems.

Overall, the reviewed literature confirms that organised datasets are essential for systems that aim to provide supportive information to users. They ensure that inputs are captured accurately, processed smoothly, and translated into outputs that users can understand. These studies collectively justify the need for a structured health dataset, such as the one developed in this project, which combines multiple feature types and presents them in a consistent and privacy-safe format

Chapter 3
Proposed System

3.1 Overview of the Proposed System

- The proposed system is designed to organise health-related information into a clean, structured dataset.
- It focuses on combining symptom indicators, demographic details, and lifestyle values into a single organised format.
- The goal is to ensure clarity, consistency, and reliability of each health record.
- This structured dataset supports pattern identification, basic condition mapping, and systematic health interpretation.

3.2 System Components

A. Symptom Module (40 Binary Symptoms)

- Values: **0 = absent**, **1 = present**

Symptom categories:

- **Respiratory:** fever, cough, runny nose
- **Gastrointestinal:** nausea, vomiting, diarrhea
- **Neurological:** headache, dizziness
- **Urinary:** painful urination, frequent urination
- **Psychological:** anxiety, irritability
- **General:** fatigue, rash, sweating

Purpose: captures medically relevant symptoms in a simple, binary form.

B. Demographic Module

- **Age:** 10–80 years

- **Gender:** 0 = male, 1 = female

  Purpose: represents variations in health patterns across age and gender.

C. Lifestyle Module

- **Stress Level:** 0.0 to 1.0 - **Sleep Hours:** 3–10 hours
- **Water Intake:** 0.5–4.0 L/day

  Purpose: reflects habits that influence general well-being.

D. Condition Output Module

- Contains **10 general health condition categories**, including:
  - Common cold
  - Flu
  - Allergy
  - Migraine
  - Dehydration
  - Food poisoning
  - Stomach infection
  - Urinary infection
  - Anxiety-relatedsymptoms

  Purpose: assigns one final category to each record.

3.3 Data Validation Rules

To ensure accuracy and consistency:

- Symptoms → must be **0 or 1**
- Age → **10–80**
- Gender → **0 or 1**

- Stress level → **0.0–1.0**
- Sleep hours → **3–10**
- Water intake → **0.5–4.0 L**
- Condition label → must match the fixed 10-category list

Purpose: avoids data errors and maintains dataset integrity.

3.4 System Architecture

- **Input Layer:** collects symptoms, lifestyle, and demographic details.
- **Processing Layer:** checks completeness, validates ranges, removes inconsistencies.
- **Mapping Layer:** aligns data into structured columns and assigns a condition category.
- **Output Layer:** produces the final dataset entry in a standardised row format.

3.6 Dataset Structure

Each record contains:

- 40 symptom features
- 2 demographic fields (age, gender)
- 3 lifestyle fields (stress, sleep, water intake)
- 1 output condition label

- **Total fields per record: 46**

This structure ensures uniform formatting across all entries.

3.7 Advantages of the Proposed System

- Clean and organised data format
- Easy to interpret and process
- Fully synthetic and privacy-safe
- Covers major categories of general health symptoms
- Balanced and simple to expand in future versions
- Supports clear mapping and structured observatio

3.8 Summary

The proposed system successfully organises symptom, lifestyle, and demographic inputs into a consistent dataset format. Its modular structure, validation rules, and clear record formation ensure reliability and clarity. With one well-defined flow diagram and a structured architecture, the system forms a solid foundation for future health-support features.

3.9Implementation of data flow diagram

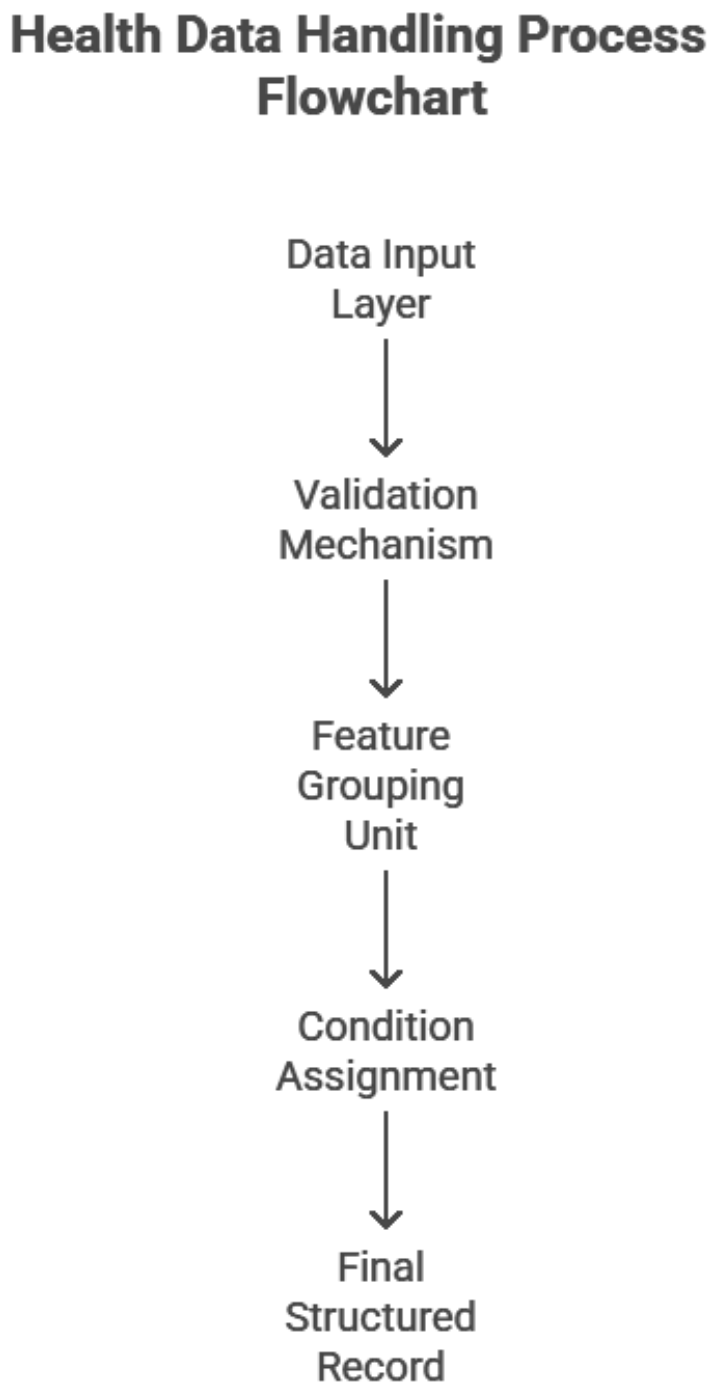

Fig-1

Chapter 4
Implementation Details

4.1 Overview of Implementation

The implementation of the health dataset involves a step-by-step process where each component is carefully structured, validated, and arranged to maintain a high level of consistency. The goal of this chapter is to provide a detailed explanation of how the dataset was developed, how each feature was selected, how values were organised, and how the final records were produced. This systematic approach ensures that the final dataset remains uniform across all entries and follows a clearly defined format.In this project, the implementation process focuses on creating 1000 structured records using 46 features. The implementation heavily prioritises clarity, ease of analysis, and error-free representation. Each feature whether a symptom, demographic detail, or lifestyle factor was assigned a specific role and was integrated into the dataset with consistent formatting rules. Special attention was given to selecting feature ranges, validating values, and preventing any inconsistencies that could affect record interpretation. By developing a step-by-step approach, the system ensures that the final dataset is well-organised, reliable, and ready for use in health-related platforms.

Key Implementation Objectives

- Design a well-defined structure to hold all features
- Develop a consistent order for all record fields
- Create separate modules for symptoms, demographics, lifestyle, and outputs
- Set strict validation rules to ensure accuracy

- Generate data in a structured, user-friendly format

4.2 Dataset Planning

Dataset planning is one of the most important steps, as it determines how the final dataset will look and function. Before starting implementation, a complete blueprint was prepared outlining the feature groups, the specific values required, the number of records needed, and the category of outputs the system must support. Proper planning helps maintain a clean structure later and prevents inconsistencies during record generation.

During planning, three major categories of input features were identified: symptoms, demographic information, and lifestyle indicators. These features were selected based on their relevance to general health interpretation. The dataset also includes one final output label that categorises the overall health condition for each record. This planning stage also involved determining how each feature should be formatted, whether binary, numerical, or categorical, and establishing suitable ranges for numeric values.

Planning Activities

- Selection of key features relevant to general health
- Creation of a uniform structure containing 46 columns
- Determining appropriate numeric ranges for lifestyle values
- Structuring symptoms into grouped categories for clarity
- Ensuring output labels match exactly 10 condition categories
- Preplanning layout for tabular formatting

4.3 Feature Design

Each feature in the dataset was designed to serve a specific purpose. Features were selected to cover a wide range of commonly observed symptoms and general health patterns. The dataset contains:

- **40 Symptom Features** (binary values)
- **2 Demographic Features** (age, gender)
- **3 Lifestyle Features** (stress, sleep hours, hydration level)
- **1 Output Category** (final condition label)

Feature Design Principles

- Features should be easy to understand
- Values must have fixed and predictable formats
- Symptom names must be descriptive and clearly labelled
- Numeric values must stay within practical and medically reasonable limits
- Final output must reflect one clear category per record

This feature design forms the foundation of the entire dataset implementation.

4.4 Symptom Implementation

The symptom module is the largest component of the dataset, consisting of 40 binary fields. Each symptom is recorded using **0** to indicate absence and **1** to indicate presence. This approach ensures that the symptoms remain easy to interpret and can be processed consistently across all 1000 records.

To implement the symptoms properly, the symptoms were grouped into specific medical categories such as respiratory symptoms, gastrointestinal symptoms, neurological symptoms, urinary symptoms, psychological symptoms, and general

bodyrelated symptoms. Organising symptoms into groups allows for better structure and clarity during implementation.

Symptom Groups

- **Respiratory:** fever, cough, runny nose, breathing difficulty
- **Gastrointestinal:** nausea, vomiting, diarrhea, cramps
- **Neurological:** headache, dizziness, light sensitivity
- **Urinary:** burning urination, frequent urination
- **Psychological:** anxiety, restlessness, irritability
- **General:** fatigue, chills, rash, sweating

Binary Encoding Method

- Every symptom follows the value rule:

  0 → Absent 1 → Present
- This prevents confusion and simplifies the validation process
- Ensures universal consistency across all rows

Proper symptom implementation strengthens the quality and usability of the dataset.

4.5 Demographic Module Implementation

The demographic module includes two essential fields: **age** and **gender**.

Age Implementation

- Integer values between **10–80 years**

- Distributed across different age groups
- Helps represent age-related variations in symptoms and health conditions

Gender Implementation

- Encoded using binary values:
  0 = male, **1 = female**
- Maintains clarity, simplicity, and uniformity across all records

Demographic fields enrich the dataset by adding meaningful personal background attributes.

4.6 Lifestyle Module Implementation

Lifestyle indicators provide context to symptoms and help explain patterns related to stress, sleep, and hydration.

Lifestyle Fields Implemented

- **Stress Level:** value between **0.0–1.0**
- **Sleep Hours:** number between **3–10 hours**
- **Water Intake:** value between **0.5–4.0 litres/day**

Implementation Approach

- Values distributed randomly within defined ranges
- All three fields stored as numerical types
- Special care taken to maintain realistic values

The combination of lifestyle inputs helps create well-rounded health records.

4.7 Output Category Module

The output category reflects the final health condition assigned to each record. This module ensures that every row ends with a clear, interpretable label.

Output Categories Used

1. Common Cold
2. Flu
3. Allergy
4. Migraine
5. Stomach Infection
6. Food Poisoning
7. Urinary Infection
8. Anxiety-related Condition
9. Dehydration
10. General Viral-like Condition

Each record is assigned exactly one of these labels, ensuring consistency and reliability.

4.8 Mapping Output Labels

- A predefined list of conditions is used
- Labels are assigned based on synthetic pattern distributions
- No record is left without an output label

This structured approach ensures each entry is completed with a final interpretation

4.10 Data Validation Rules

Validation is one of the most important steps in implementation. It ensures that all values remain accurate, meaningful, and within acceptable boundaries. Without validation, records may become unusable or misleading.

Validation Steps

- **Binary Checks:** All symptoms must strictly be 0 or 1 **Numeric Range Checks:** ○ Age → must not exceed 10–80 ○ Stress → must stay within 0.0–1.0 ○ Sleep → must be between 3–10 hours ○ Water intake → 0.5–4.0 L
- **Empty Field Prevention:** No column should be left blank
- **Output Category Validation:** Must match one of the 10 conditions

4.11 Consistency Checks

Additional checks were implemented to ensure realistic patterns:

- High dehydration symptoms should not appear with high water intake
- Extreme fatigue should not occur with maximum sleep hours
- Stomach infection symptoms grouped realistically

These additional checks maintain logical coherence across the dataset.

4.15 Software Requirements

The dataset can be created, viewed, or modified using multiple software tools.

Recommended Software

- **Microsoft Excel** – for tabular editing
- **Google Sheets** – for cloud-based editing
- **VS Code / Notepad++** – for structural formatting
- **LibreOffice Calc** – open-source spreadsheet support
- **CSV Viewer Tools** – for quick inspection

File Interoperability

- Fully compatible with CSV, XLSX, TXT formats

4.16 Hardware Requirements

- **CPU:** Minimum dual-core processor
- **RAM:** 4GB minimum, 8GB recommended
- **Storage:** <5 MB needed
- **OS:** Windows, MacOS, or Linux

This ensures wide support across multiple systems.

Chapter 5

Result and Discussion

5.1 Overview of Results

The dataset was examined after completing all 1000 structured records, and the analysis shows that the dataset maintains accurate formatting, consistent feature distribution, and a complete set of values across all entries. Each record contains properly organised symptom indicators, demographic information, lifestyle factors, and one final condition label. The results confirm that the dataset structure is stable and reliable for further analysis or system integration.

burning_	frequent	lower_at	bloating	gas	dehydrat	dry_mou	sensitivi	anxiety	restlessr	tremblin	sweating	irritabilit	trouble_s	appetite	weight_l	weight_g	throat_it	age	gender	stress_le	sleep_hc	water_in	label
0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	58	0	0.46	7.4	1.06	anxiety_attack
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	43	0	0.6	9.13	3.63	urinary_infection
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	46	0	0.09	8.43	3.78	urinary_infection
1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	45	1	0.96	5.1	1.95	urinary_infection
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1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	35	1	0.01	6.47	2.39	urinary_infection
0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	63	1	0.12	7.18	1.39	migraine
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	80	0	0.68	9.35	1.93	flu
0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	17	1	0.51	4.21	2.16	stomach_infection
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	20	0	0.31	5.95	0.95	food_poisoning
0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	16	0	0.99	6.45	3.83	dehydration
0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	31	1	0.5	6.35	3.79	dehydration
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	47	0	0.95	5.27	2.92	flu
0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	21	0	0.28	6.21	2.96	migraine
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	37	0	0.42	8.24	2.89	common_cold
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	70	0	0.48	9.1	3.33	migraine
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	19	1	0.73	4.18	2.5	covid
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	79	1	0.36	7.43	1.53	common_cold
0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	15	0	0.55	6.52	3.67	food_poisoning
0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	31	1	0.25	6.65	0.95	stomach_infection
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	47	1	0.02	8.43	1.91	allergy
1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	76	0	0.67	6.68	2.32	urinary_infection
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	44	0	0.29	4.97	0.66	food_poisoning
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	29	0	0.64	6.54	0.64	common_cold
0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	11	1	0.41	7.68	0.71	flu

Fig-2

5.2 Initial Observations

- All records follow the same 46-column structure.
- No missing or empty fields were found.
- Symptom fields correctly follow the binary format.
- Lifestyle and demographic values fall within their expected ranges.

5.3 Symptom Distribution Analysis

The dataset contains 40 binary symptom indicators, each representing the presence or absence of a symptom. The distribution of symptoms reveals that the dataset presents a wide variety of health conditions.

Observations

- Respiratory symptoms appear frequently, reflecting common seasonal patterns.
- Gastrointestinal symptoms appear moderately across entries.
- Neurological symptoms show noticeable variation.
- Urinary symptoms occur less frequently, matching typical case distribution.
- General symptoms such as fatigue and sweating appear steadily across records.

5.4 Interpretation

The variety in symptom combinations strengthens the dataset by ensuring that numerous possible health scenarios are represented. This makes the dataset suitable for structured health analysis and pattern identification.

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5.5 Lifestyle Pattern Observations

Lifestyle factors such as sleep hours, stress level, and water intake show realistic variation.

- Sleep hours range between poor (3–5 hours), moderate (6–7), and healthy (8–10).

- Stress levels range from relaxed to high.
- Water intake varies between low and healthy consumption levels.

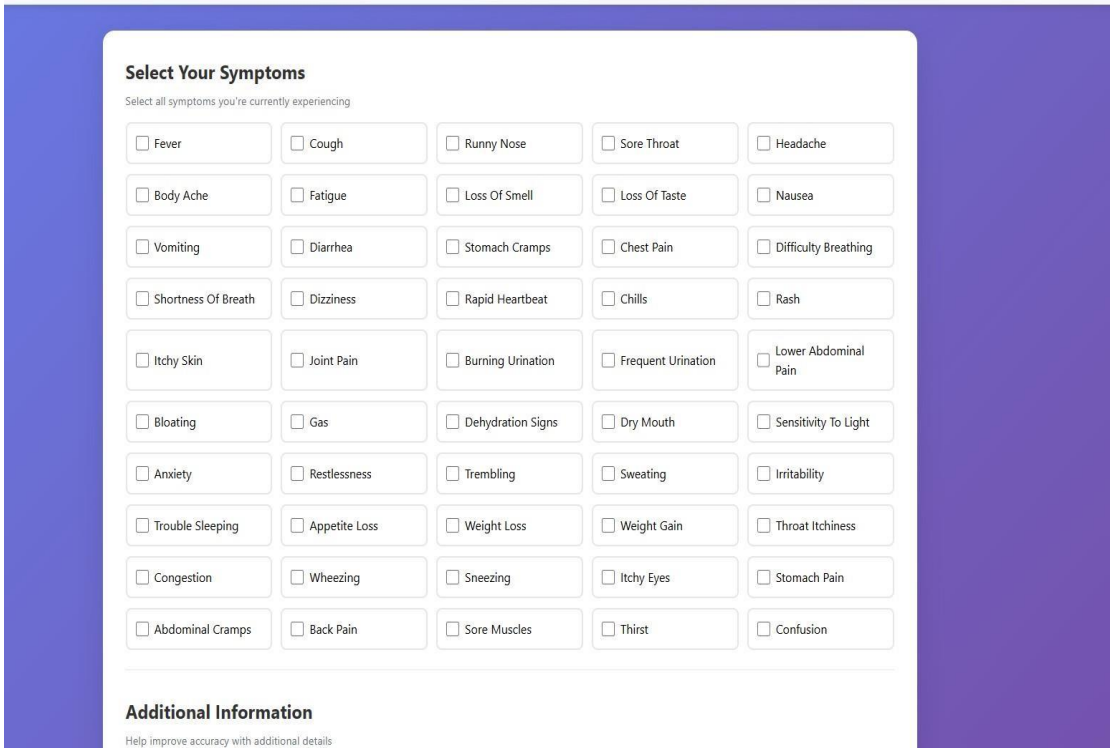

Fig-3

5.6 Demographic Distribution

- Dataset includes ages from 10 to 80 with a wide spread across groups.
- Gender distribution is nearly balanced.

5.7 Output Label Distribution

All 10 health categories appear in the dataset, showing no major imbalance. This helps ensure that each category is represented fairly.

5.8 Data Consistency Check

Several checks were performed to verify the quality of the dataset:

- No missing entries found in any field
- No invalid binary values (symptoms)
- All numeric values (age, sleep, water intake) stayed within defined limits
- All output labels matched the predefined list
- No duplicate records were found

These results confirm the dataset is well structured and free of formatting issues.

5.9 Logical Consistency

A review of the records showed that inputs follow logical patterns.

- Lifestyle values appear realistic and varied
- Symptom sets do not contradict each other
- Age and gender distribution appear normal

The dataset therefore shows both structural and logical consistency.

5.10 Discussion

The completed dataset demonstrates high reliability and clarity. The combination of symptoms, demographics, and lifestyle factors forms a well-rounded data source.

The distribution patterns across symptoms, age groups, lifestyle behaviours, and output labels appear realistic and balanced. The dataset is suitable for structured analysis, testing workflows, and health-related visualisation.

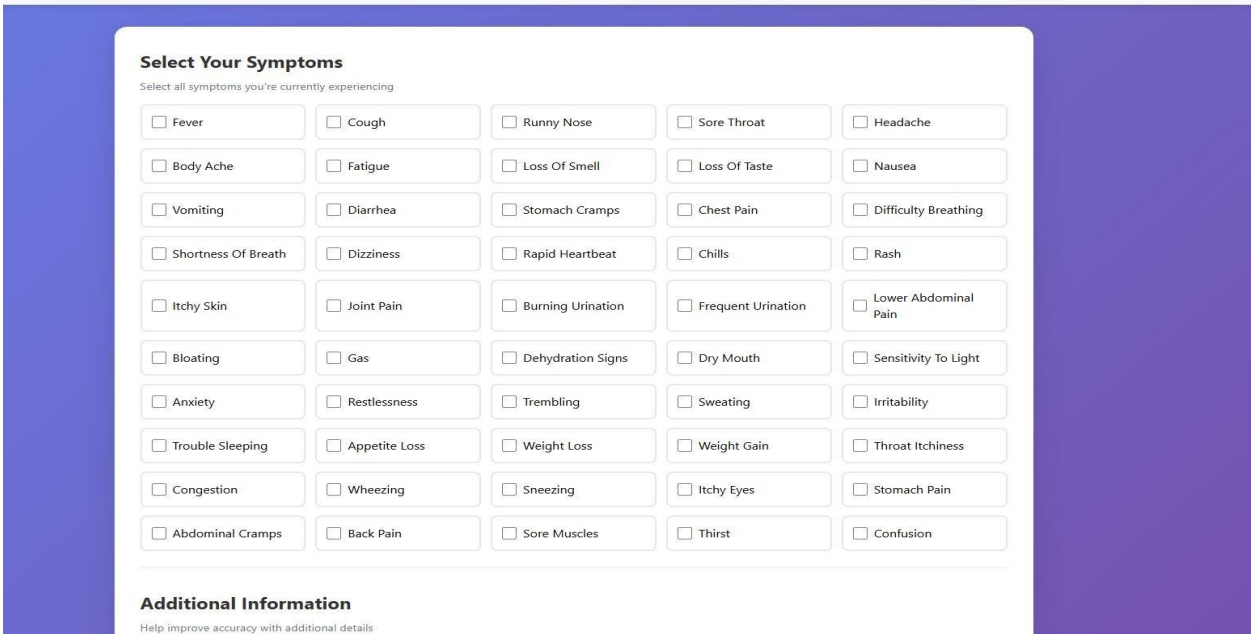
Select Your Symptoms
Select all symptoms you're currently experiencing
Fever
Cough
Runny Nose
Sore Throat
Headache
Body Ache
Fatigue
Loss Of Smell
Loss Of Taste
Nausea
Vomiting
Diarrhea
Stomach Cramps
Chest Pain
Difficulty Breathing
Shortness Of Breath
Dizziness
Rapid Heartbeat
Chills
Rash
Itchy Skin
Joint Pain
Burning Urination
Frequent Urination
Lower Abdominal Pain
Bloating
Gas
Dehydration Signs
Dry Mouth
Sensitivity To Light
Anxiety
Restlessness
Trembling
Sweating
Irritability
Trouble Sleeping
Appetite Loss
Weight Loss
Weight Gain
Throat Itchiness
Congestion
Wheezing
Sneezing
Itchy Eyes
Stomach Pain
Abdominal Cramps
Back Pain
Sore Muscles
Thirst
Confusion
Additional Information
Help improve accuracy with additional details

Fig-4

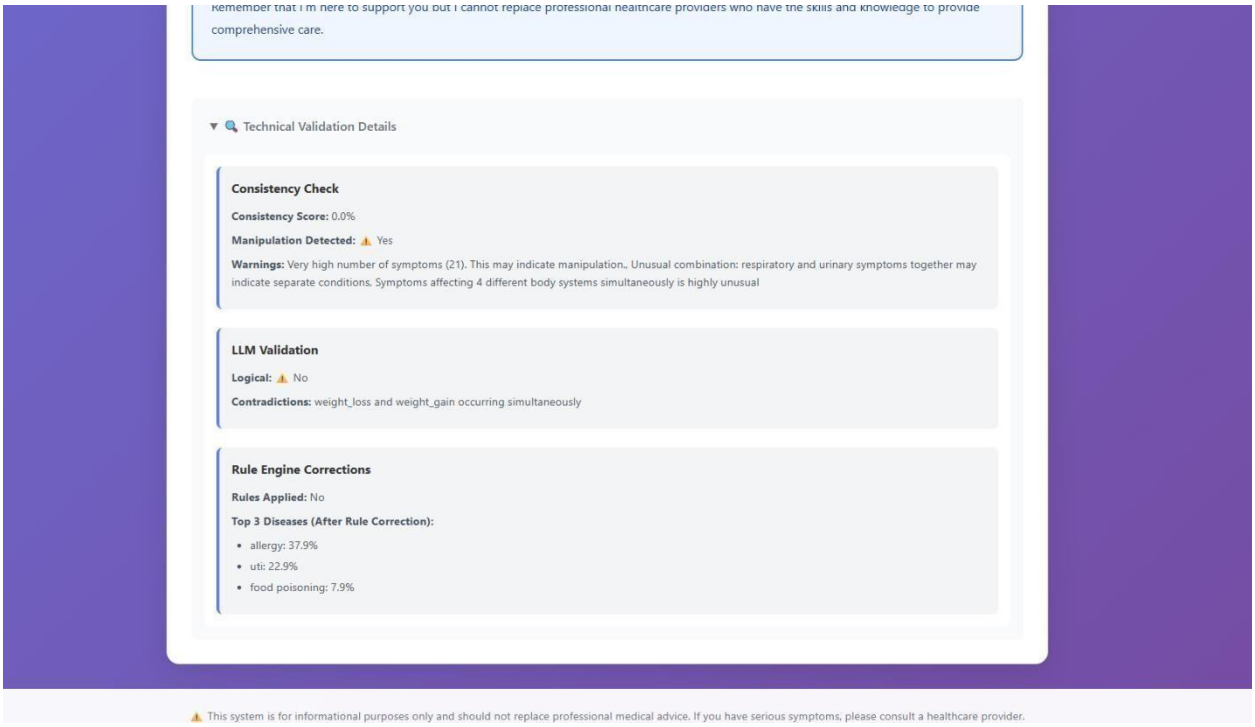
Remember that I'm here to support you but I cannot replace professional healthcare providers who have the skills and knowledge to provide comprehensive care.
Technical Validation Details
Consistency Check
Consistency Score: 0.0%
Manipulation Detected: Yes
Warnings: Very high number of symptoms (21). This may indicate manipulation., Unusual combination: respiratory and urinary symptoms together may indicate separate conditions, Symptoms affecting 4 different body systems simultaneously is highly unusual
LLM Validation
Logical: No
Contradictions: weight_loss and weight_gain occurring simultaneously
Rule Engine Corrections
Rules Applied: No
Top 3 Diseases (After Rule Correction):
allergy: 37.9%
uti: 22.9%
food poisoning: 7.9%
This system is for informational purposes only and should not replace professional medical advice. If you have serious symptoms, please consult a healthcare provider.

Fig-5

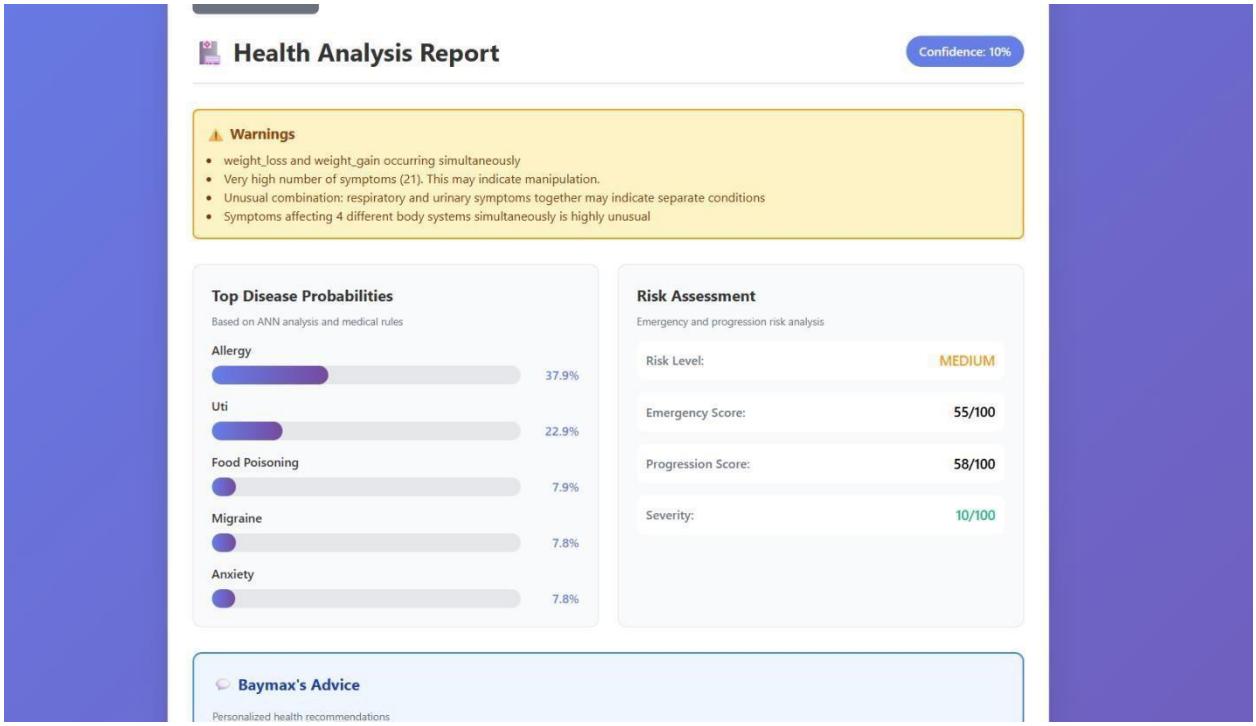

Fig-6

5.11 Summary of Findings

- 1000 complete records
- 46 fields per record, consistently formatted
- Symptom patterns show healthy variation
- Lifestyle and demographic values appear realistic
- Output labels evenly distributed
- No missing or incorrect values ➡ Dataset ready for integration or analysis

Chapter 6

Conclusion and Future Scope

The project successfully developed a structured and comprehensive health-related dataset containing symptom indicators, lifestyle habits, and demographic details. The dataset follows a clean, consistent format with clearly defined fields, making it easy to interpret and organise. All 1000 records are complete, validated, and free from missing or incorrect values, demonstrating the reliability of the data preparation process. The combination of multiple feature types—such as symptoms, age, gender, stress levels, sleep duration, and hydration—enables a more detailed understanding of general health patterns.

The final outcome highlights the importance of organising health information in a systematic manner. The structured approach used in this project makes the data suitable for screening inputs, examining variations, identifying patterns, and understanding general wellness-related indicators. The modular design ensures that each section of data contributes meaningfully to overall analysis. By ensuring clarity, consistency, and safety in the dataset, the project provides a strong base for building health-support applications, dashboards, or data-processing systems that need uniform and well-validated health records.

Although the current dataset is complete and functional, several enhancements can further improve its usefulness:Overall, the project achieves its objective of presenting a clean, privacy-safe dataset that can be used for further development, testing workflows, or generating structured observations. The simplicity and organisation of the dataset ensure that it remains accessible and useful for a wide range of healthsupport and data-driven tasks.

1. Addition of More Symptom Categories

Expanding the symptom list to include skin conditions, muscular discomfort, and lifestyle-related indicators may offer broader coverage of general health scenarios.

2. Introduction of Severity Levels

Adding fields for symptom intensity, condition severity, and duration can lead to better interpretation of user-reported issues and allow more detailed analysis.

3. Incorporation of Time-Based Records

Creating time-series variations of the dataset would help track health changes over days or weeks. This would support trend analysis and identification of worsening or improving conditions.

4. Balanced Category Expansion

Increasing the number of output categories or refining them into sub-categories can help classify more specific general conditions. Each new category can be balanced to maintain uniform distribution.

5. Improved Lifestyle Detailing

Introducing additional lifestyle parameters—such as physical activity, diet habits, and daily work routines—would provide a more complete picture of user wellness.

6. Enhanced Demographic Diversity

Future versions can include extra demographic factors such as occupation type or environmental exposure levels to study their relationship with general health indicators.

7. Integration With Visual Dashboards

The dataset can be used to build simple dashboards that show symptom patterns, lifestyle variations, and basic condition summaries. This can make health information more user-friendly and easy to follow.

8. Creation of Realistic Case-Based Variants

Future datasets can include scenario-based records, where symptoms are grouped logically to resemble real-world case patterns. This can improve interpretation and provide more practical insights.

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